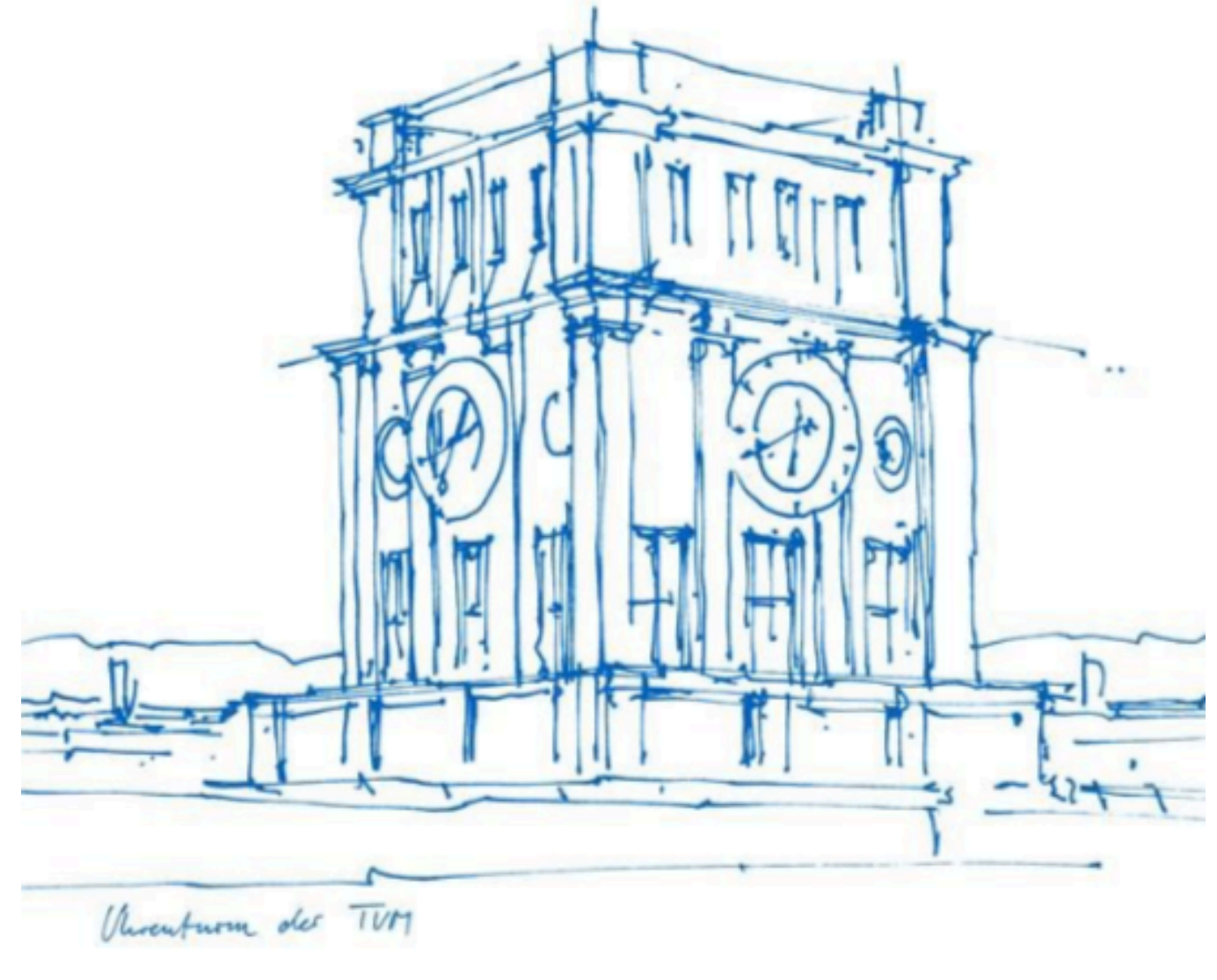


Using LLM for Automated Mobile App Usability / Accessibility Testing

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Agenda

- Introduction
- Motivation
- Overview of Reviewed Approaches
- Results
- Conclusion

Introduction

- **usability** – “a concept that essentially refers to how easy it is for users to learn a system, how efficient they can be once they have learned it, and how enjoyable it is to use it”
- **accessibility**– being designed and developed so that people with disabilities can use them

How are usability/accessability traditionally tested?

- manual testing with real users
- user experience expert evaluation

advantages

- realistic
- context aware

disadvantages

- expensive
- time consuming
- limited coverage

Motivation

That is why automated UI testing started to develop

Traditional automated usability/accessability testing:

- checks only predefined rules that are unambiguous
- does not check the meaning



need for still automated testing but with human-like understanding

LLM

At what stages do the LLMs assist with automation?

is the UI accessible?

how usable is the implemented UI?

Implementation Stage



Design Stage

how usable is the UI's design?

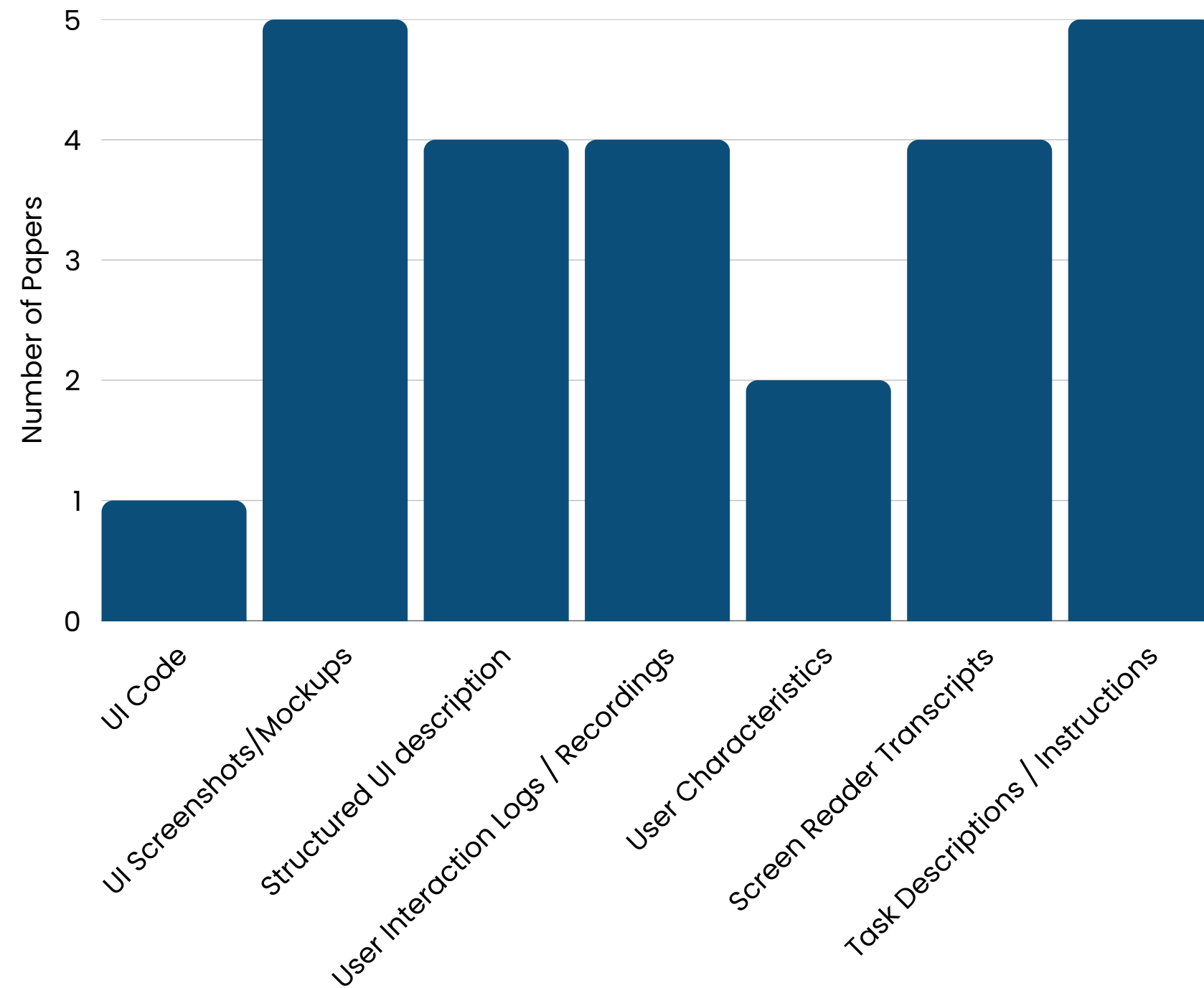
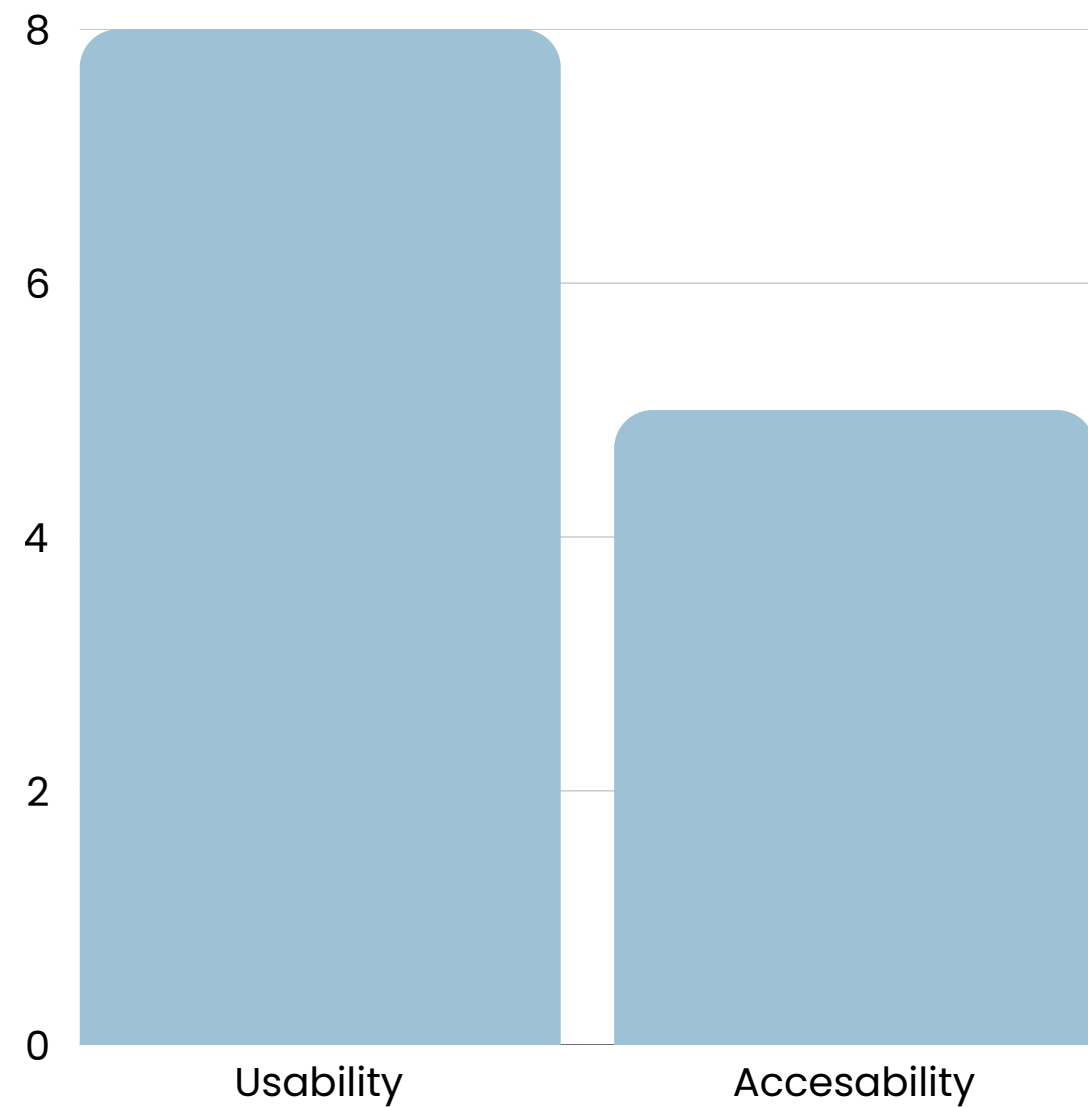
Test Report Stage

can we automate generating test reports?

Automating usability/accessability testing with LLMs



What is the LLMs s' input?



Automating usability/accessability testing with LLMs

Static Approaches

- app is not run during testing

Example: LLM is given UI screens, code and a question: Identify Nielsen heuristics violations

Dynamic Approaches

- app is run and interacted with

LLM gets the UI code or the transcription of the Screen Reader and the goal, based on that decides what to do

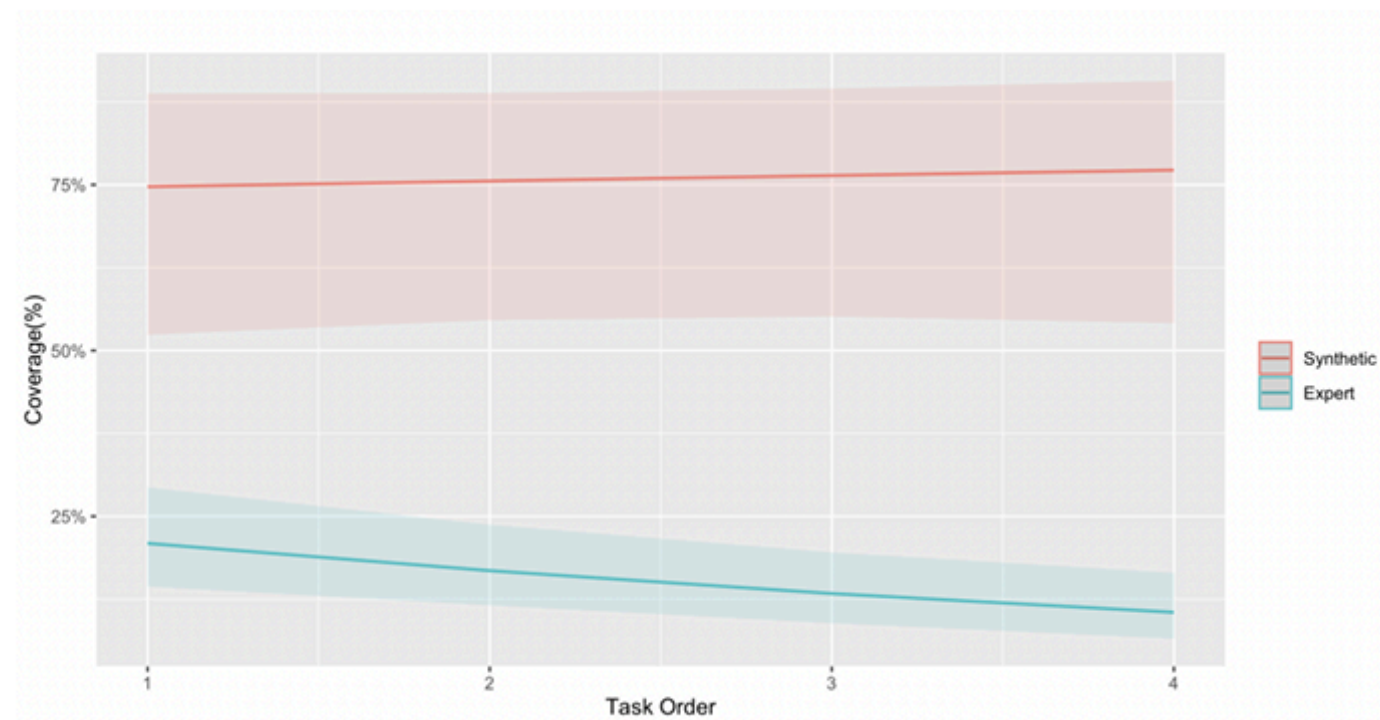
The UI gets updated and LLM gets information about the updated view

Results

LLMs can find most of the usability and accessibility issues found by the human testers

advantages

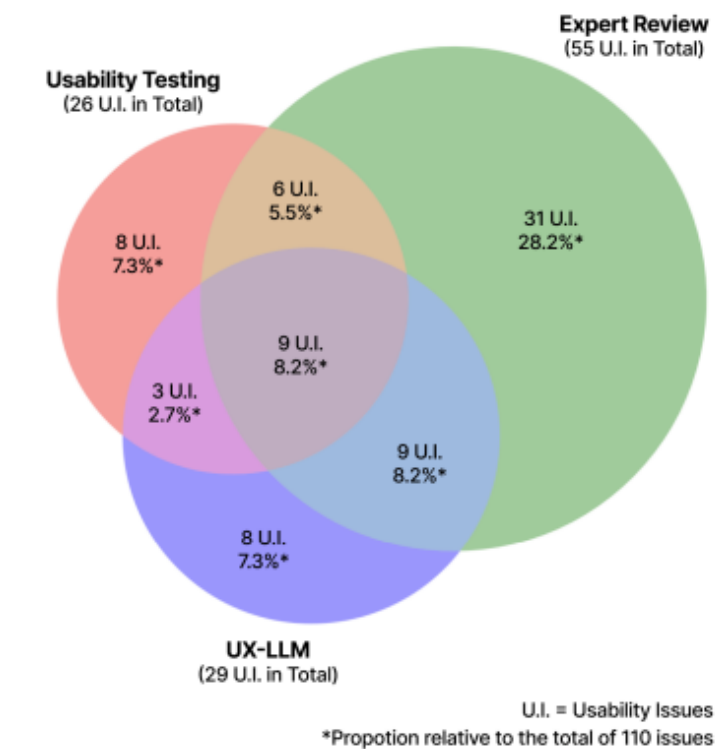
- consistent performance across tasks
- strong at layout / visual consistency issues
- fast



Paper: Synthetic Heuristic Evaluation: A Comparison between AI- and Human-Powered Usability Evaluation

disadvantages

- worse at cross-screen violation
- static miss the dynamic content



Paper: Does GenAI Make Usability Testing Obsolete?

Conclusion



Common Conclusion:

- Automation by LLMs: way to reduce costs, reduce time but not a way to replace manual testing
- Proposed solution: use both manual and LLM automated solutions
 - static approaches → good for early feedback, cheap, scalable
 - dynamic approaches → expensive, harder to scale
- accessibility testing mostly focus on screenreaders, other not broadly tested for LLM automation

Thank you for your attention!